INCH-POUND

MIL-R-29533(EC)
14 July 1989

MILITARY SPECIFICATION

RADIO SET, AN/URC-( )

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers minimum performance and safety requirements for a multichannel Radio Set (Transceiver), AN/URC-( ), utilizing frequency modulation in the 156 megahertz (MHz) through 162 MHz frequency range. The Radio Set, hereinafter referred to as the equipment, shall provide voice communications subject to the Federal Communications Commission (FCC) rules and regulations associated with shipboard stations in the Maritime Service.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR PMW 152), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5820

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment, And Systems, Requirements For
MIL-E-16400G	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment Accessories, And Provisioned Items (Repair Parts): Packaging Of
MIL-M-38510	Microcircuits, General Specification For

STANDARDS

MILITARY

MIL-STD-108	Definitions Of Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-242, Part II	Electronic Equipment Parts Selected Standards, Radio Frequency Components
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-471	Maintainability Verification/Demonstration/ Evaluation
MIL-STD-781C	Reliability Design Qualification And Production Tests: Exponential Distribution
MIL-STD-810C	Environmental Test Methods
MIL-STD-965	Parts Control Program

Source: https://assist.dla.mil -- Downloaded: 202640-21T00:00Z
Check the source to verify that this is the currentversion before use.

HANDBOOKS

MILITARY

MIL-HDBK-175	Microelectronic Device Data Handbook
MIL-HDBK-781	Reliability Test Methods, Plans, And Environmental For Engineering Development *Qualification, And Production*

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

FEDERAL

FCC

FCC Rules, Part 83

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

MILITARY

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

ELEX 0967-LP-464-8010 Technical Manual For AN/URC-80(V)5

(Copies of specifications, standards, handbooks, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA RS-152-C-88	Minimum Standards For Land-Mobile Communications FM Or PM Transmitters, 25-866 MHz
EIA RS-204-C-82	Minimum Standards For Land-Mobile Communications FM Or PM Receivers, 25-947 MHz

(Application for copies should be addressed to the Electronic Industries Association, Engineering Department, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The equipment, including parts, materials, and processes, shall comply with the requirements of MIL-E-16400G and FCC rules and regulations, to the extent specified herein. The equipment shall be a very high frequency (VHF) frequency modulated (FM) bridge-to-bridge maritime communications set for shipboard, and small boats and craft installations. The equipment shall use microcircuit techniques and solid-state technology to the maximum extent practical.

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Description. The equipment shall contain a receiver/transmitter (R/T) capable of providing 55 selectable transmitting channels and 61 selectable receiving channels. The equipment shall be designed for both simplex operation (f1/f1) (see 6.5.1) and half duplex operation (f1/f2) (see 6.5.2) in accordance with the channelization requirements of 3.1.10 over the frequency range of 156.025 MHz to 162.550 MHz. In addition to a selectable R/T section, the equipment shall have one operated guard receiver fixed-tuned to 156.650 MHz (channel-13). The equipment shall provide radio emergency channel selection, high degree of reliability, multiple remote control capability, key pad

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

tuning, and a built-in test (BIT) (see 6.5.5) system capable of detecting faults in every module in the R/T unit. Channel selection shall be accomplished by keying the relevant number of the key pad for rapid selection of the emergency frequency channel (channel-16). The equipment shall operate from a single antenna with both receivers operating continuously, except during transmissions.

3.1.3 Design and construction. Design and construction of the equipment shall conform to the requirements of MIL-E-16400G to the extent specified herein. The equipment shall be designed and fabricated using solid-state electronic circuitry throughout without the use of vacuum tubes and electromechanical tuning for frequency selection (see NAVELEX 0967-LP-464-8010 for guidance). Design and construction shall be as specified in 3.1.3.1 through 3.1.6.

3.1.3.1 Parts, materials, and processes. Parts, materials, and processes for the equipment shall be in accordance with the Parts, materials and processes paragraph of MIL-E-16400G, except as specified in 3.1.3.1.1 through 3.1.3.1.1.6.

3.1.3.1.1 Parts. Parts shall be as specified in 3.1.3.1.1.1 through 3.1.3.1.1.6.

3.1.3.1.1.1 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.1.3.1.1.2 Microelectronic modular assemblies. All microelectronic assemblies and devices employed in the design of the equipment shall comply with Class B requirements of MIL-M-38510. MIL-HDBK-175 shall be used as guidance in the selection and application of all microelectronic devices used in the equipment.

3.1.3.1.1.3 Module sockets. All module sockets shall be readily accessible for easy removal and replacement.

3.1.3.1.1.4 Printed wiring boards (PWBs). PWBs shall conform to MIL-STD-454, Requirement 17.

3.1.3.1.1.5 Printed circuit (PC) cards. PC cards shall conform to the requirements of 3.1.3.1 with respect to design, connectors, components, and assembly (see 6.5.4 and 6.5.9). With power applied to the equipment, removal or insertion of PC assemblies shall not damage the assembly or any other part of the equipment. Each assembly shall be keyed to prevent incorrect insertion. PC cards containing electrostatic discharge sensitive (ESDS) (see 6.5.7) components shall be prominently marked with the sensitive electronic device symbol in accordance with MIL-STD-129, in a position readily visible when installed. Whenever possible, ESDS parts (see 6.5.8) should be located on PC cards so that pins of

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

these parts are not directly connected to the input or output of the PC card. Card edge connectors shall not be used.

3.1.3.1.1.6 Electronic components. Electronic components (digital or solid-state) shall conform to the requirements of MIL-STD-242, Part 11, except as otherwise specified herein. Electronic components used in the equipment design shall be available from more than one commercial source, unless otherwise approved by the procuring activity.

3.1.4 Circuit test points. Circuit test points shall be provided which comply with the criteria specified in MIL-STD-415. The types of tests specified in a through h shall be considered for determining test points:

a. Signal quality
b. Circuit performance
c. Calibration
d. System alignment
e. Installation and checkout
f. Planned maintenance system tests
g. Line replaceable unit (LRU) fault location
h. Shop replaceable part (SRP) fault isolation

Test points for calibration shall be provided external to the unit under test, except where such provision would affect the calibration. Test points shall be provided with the proper interface fittings and matching circuits to connect the selected support test equipment.

3.1.5 Operational control. The R/T unit shall be capable of being remotely controlled via cable to a maximum distance of 2286 centimeters (cm) (75 feet (ft)) by a control unit which performs the functions provided by the controls and switches of 3.2.4.1.

3.1.6 Equipment readiness. The equipment shall be complete, ready for installation and operation, and shall include installation brackets, hardware, and mating connectors. The audio output connector and the input connector of the R/T unit shall each be capable of mating with an N type connector receptacle. The equipment shall have, as a minimum, the basic items specified in a through f:

a. R/T unit (including):
    1. Receivers (2)
    2. Transmitter (1)
    3. Power supply
b. Control head front panel
c. Loudspeaker

Source: https://assist.dla.mil -- Downloaded: 202610-21T00:00Z
Check the source to verify that this is the current version before use.

d. Antenna
e. 2286 cm (75 ft) cable (control unit to R/T unit)
f. Handset

3.1.7 FCC rules. The equipment shall have been type-accepted and certified under the FCC rules and regulations governing stations on shipboard in the Maritime Service (Part 83) for operation after 1 January 1973.

3.1.8 System deviation and channel spacing. The equipment shall be designed for narrowband operation. Narrowband operation shall be in operation with radio frequency (RF) channels spaced 25 kilohertz (kHz) apart with a maximum frequency deviation of ±5 kHz at 100 percent modulation.

3.1.9 Emission characteristics. Emission characteristics of 16F3 for narrowband operation shall be in accordance with FCC Rules, Part 83.

3.1.10 Channelization. The guard receiver shall be single channel, fixed tuned, preprogrammed, and aligned to operate on 156.650 MHz prior to shipment. The main transmitter and receiver shall be preprogrammed and aligned to operate on the frequencies specified in TABLE I. The radio set shall be capable of operating in both the US and International modes of operation. The transmitter section shall be inhibited on the channels specified in TABLE I.

TABLE I. Narrowband frequencies.

Channel	Transmitting frequency (MHz)	Receiving frequency (MHz)	
		US mode	International mode
00	Inhibit	160.600	
01	156.050	160.650	
02	156.100	160.700	
03	156.150	160.750	
04	156.200	160.800	
05	156.250	160.850	
06	156.300	156.300	
07	156.350	156.350	160.950
08	156.400	156.400	
09	156.450	156.450	
10	156.500	156.500	
11	156.550	156.550	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

TABLE I. Narrowband frequencies - Continued.

Channel	Transmitting frequency (MHz)	Receiving frequency (MHz)	
		US mode	International mode
12	156.600	156.600	
13	156.650	156.650	
14	156.700	156.700	
15	156.750	156.750	
16	156.800	156.800	
17	156.850	156.850	
18	156.900	156.900	156.500
19	156.950	156.950	161.550
20	157.000	161.600	
21	157.050	157.050	161.650
22	157.100	157.100	161.700
23	157.150	157.150	161.750
24	157.200	161.800	
25	157.250	161.850	
26	157.300	161.900	
27	157.350	161.950	
28	157.400	162.000	
29	Inhibit	157.450	162.050
60	156.025	160.625	
61	156.075	160.675	
62	156.125	160.725	
63	156.175	160.775	
64	156.225	160.825	
65	156.275	156.275	160.875
66	156.325	156.325	160.925
67	156.375	156.375	
68	156.425	156.425	
69	156.475	156.475	
70	156.525	156.525	
71	156.575	156.575	
72	156.625	156.625	
73	156.675	156.675	
74	156.725	156.725	
75	Inhibit	156.775	
76	Inhibit	156.825	
77	156.875	156.875	
78	156.925	156.925	161.525

Source: https://assist.dla.mil -- Downloaded: 202610-21T00:00Z
Check the source to verify that this is the current version before use.

TABLE I. Narrowband frequencies - Continued.

Channel	Transmitting frequency (MHz)	Receiving frequency (MHz)	
		US mode	International mode
79	156.975	156.975	161.575
80	157.025	157.025	161.625
81	157.075	157.075	161.675
82	157.125	157.125	161.725
83	157.175	157.175	161.775
84	157.225	161.825	
85	157.275	161.875	
86	157.325	161.925	
87	157.375	161.975	
88	157.425	157.425	
89	Inhibit	157.475	
W	Inhibit	162.550	

3.1.11 Frequency synthesis. Generation of all channels shall be accomplished by frequency synthesis. Changing of channels shall not require the removal or insertion of crystals, other components, or realignment.

3.1.12 Antenna overload. The transmitter final amplifier shall not be damaged under variable load conditions, such as, grounding, partial grounding, or disconnecting of the antenna. An input signal (10 volts root-mean-square (rms) at any frequency at the antenna jack) shall not damage the receiver.

3.1.13 Physical characteristics. Physical characteristics shall be as specified in 3.1.13.1 through 3.1.13.3.

3.1.13.1 Dimensions. The overall dimensions of the R/T shall not exceed 30.48 cm (12 inches (in.)) wide by 48.26 cm (19 in.) deep by 20.32 cm (8.0 in.) high. The overall dimensions of the control head front panel shall not exceed 37.97 cm (14.95 in.) wide by 18.54 cm (7.3 in.) deep by 19.05 cm (7.5 in.) high.

3.1.13.2 Weight. The weight of the equipment (including the R/T and control head front panel) shall not exceed 18.14 kilograms (40 pounds).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

3.1.13.3 Modular construction. The equipment shall use modular construction to the maximum extent practicable.

3.1.14 Built-in test equipment (BITE) (see 6.5.6). BITE shall be implemented within the equipment which shall permit the isolation of a fault to a single module or assembly. All BITE controls and indicators shall be mounted on the front panel.

3.1.15 Illumination of operating controls. All equipment controls shall be illuminated.

3.2 Electrical requirements. Electrical requirements shall be as specified in 3.2.1 through 3.2.6.

3.2.1 Alternating current (AC) power source. The equipment shall be capable of operating from a 115-volts alternating current (VAC), 60-hertz (Hz) ±3 Hz single-phase power source.

3.2.1.1 Direct current (DC) power source. The equipment shall be capable of operating from a 24-volts direct current nominal ±10 percent power source.

3.2.1.2 Power supply voltage range. Equipment performance shall conform to the requirements of the Power supply voltage range minimum standard paragraph of EIA RS-152-C-88 for transmitters, and EIA RS-204-C-82 for receivers.

3.2.1.3 AC power cable. No AC power cable shall be supplied, but AC mating connector to R/T unit shall be provided.

3.2.1.4 Reverse polarity protection. The R/T shall include means to protect the equipment against accidental reversal of polarity when connected to the DC power source.

3.2.1.5 Fuses. All AC and DC electrical power inputs to the R/T shall be protected by fast-blow type fuses of adequate rating.

3.2.1.6 Electrical power connectors. Separate connectors shall be provided for AC and DC power connectors.

3.2.1.7 Bonding and grounding. Bonding and grounding shall be in accordance with the requirements of the Bonding and grounding paragraph of MIL-E-16400G.

3.2.1.8 Leakage current. The leakage current of the equipment shall not exceed 5 milliamperes (mA) at 115-VAC, 60-Hz, single-phase source power.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

3.2.1.9 Safety. The requirements of the Safety criteria paragraph of MIL-E-16400G shall be incorporated in the design of the equipment to preclude personnel from accidentally coming into contact with hazardous electrical potentials.

3.2.1.9.1 Equipment power connectors. Equipment power connectors for 115-VAC input power, and interconnecting AC power cable connectors between equipment, shall have only three pins designated A, B, and C. Pin assignment and color code shall be as specified in TABLE II. The black-and-white color code shall be maintained from the input connector pins to all components having the same voltage and frequency as the input power.

TABLE II. Pin assignment and color code.

Pin designation	Conductor assignment	Color code
A	115-VAC return	White
B	Safety ground	Green
C	115-VAC high (hot)	Black

3.2.1.9.2 Critical controls. Critical controls, the accidental activation of which may cause damage to equipment, injury to personnel, or degradation of system function, shall be designed and located so that they are not susceptible to being moved accidentally.

3.2.2 Transmitter performance requirements. The transmitter shall conform to all requirements specified herein under the Standard test conditions paragraph of EIA RS-152-C-88 for narrowband operation on all RF channels throughout the channelization specified (see 3.1.10).

3.2.2.1 Rated RF power output. The transmitter shall conform to the RF power restrictions specified in the FCC Rules. Power output shall be constant between 20 watts (W) to 25 W reducible to 0.75 W to 1.0 W by a power level switch. The transmitter shall be designed for alignment and operation with load impedances corresponding to voltage-standing-wave-ratios from 1.0:1 to a maximum of 3:1 with respect to 50 ohms. The transmitter shall have an output impedance of 50 ohms. Carrier power shall not drop below maximum carrier power output during a duty cycle of one minute ON and one minute OFF for 8 hours.

3.2.2.2 Conducted and radiated spurious emissions. The conducted and radiated spurious emissions shall be greater than 90 decibels (dB) below the carrier.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

3.2.2.3 Audio frequency harmonic distortion. The audio frequency harmonic distortion shall not exceed 5 percent.

3.2.2.4 Audio frequency response. The audio frequency response shall be in accordance with EIA RS-152-C-88.

3.2.2.5 FM hum and noise level. The FM hum and noise level shall be at least 40 dB below standard test modulation.

3.2.2.6 Modulation limiting. The modulation limiting shall be in accordance with EIA RS-152-C-88.

3.2.2.7 Carrier frequency stability. The carrier frequency stability shall be within ±0.0005 percent of the assigned carrier frequency over a minimum 10-hour period when tested to the limits specified in 3.4.1 and 3.4.2.

3.2.2.8 Amplitude modulation (AM) hum and noise. The requirement limit for AM hum and noise shall be in accordance with EIA RS-152-C-88.

3.2.2.9 Transmitter sideband spectrum. The requirement limit for the transmitter sideband spectrum shall be in accordance with EIA RS-152-C-88.

3.2.2.10 Audio inputs. Audio inputs shall be as specified in a and b:

a. Local inputs. The transmitter shall be capable of operating from audio signals generated by a standard microphone, or a field handset.

b. Remote audio input. The transmitter shall be capable of operating from remote audio inputs of -20 decibels referred to one milliwatt (dBm) to +10 dBm, 600 ohms balanced line.

3.2.2.11 Sidetone. Sidetone capability shall be provided to the equipment. The control head front panel shall enable the microphone audio to be applied to the 5-W audio receiver amplifier. Sidetone capability should allow bridge personnel to hear two-way communications. The level of the sidetone shall be adjustable up to 1 W, maximum output.

3.2.3 Receiver performance requirements. Simultaneous operation from the radio set antenna of the two receivers shall not cause degradation in performance of either receiver below the performance specified herein. The input impedance of each receiver shall be 50 ohms. Minimum standards specified in EIA RS-204-C-82 shall be disregarded, unless otherwise specified in 3.2.3.1 through 3.2.3.15 with the receiver conforming to the requirements specified herein. The receiver shall conform to the requirements for narrowband operation in accordance with the channelization requirements of 3.1.10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

3.2.3.1 Usable sensitivity. The usable sensitivity shall be 1.0 microvolt (μV) (open circuit) maximum.

3.2.3.2 Quieting sensitivity. The quieting sensitivity shall be 1.0 μV (open circuit) maximum.

3.2.3.3 Audio squelch sensitivity. The threshold sensitivity shall be 0.75 μV (open circuit) maximum.

3.2.3.4 Modulation acceptance bandwidth. The modulation acceptance bandwidth shall be greater than ±7 kHz.

3.2.3.5 Adjacent channel selectivity and desensitization. The adjacent channel sensitivity and desensitization shall be 70 dB minimum.

3.2.3.6 Spurious response attenuation. Spurious response attenuation shall be a minimum of 100 dB.

3.2.3.7 Intermodulation spurious attenuation. The intermodulation spurious attenuation limit shall be as specified in the FCC Rules, Part 83.

3.2.3.8 Audio power output. The maximum audio power output shall be a minimum of 4.0 W at not more than 5 percent distortion.

3.2.3.9 Audio frequency response. The audio frequency response shall be in accordance with EIA RS-204-C-82.

3.2.3.10 Hum and noise. The hum and noise ratio shall be in accordance with EIA RS-204-C-82.

3.2.3.11 Undesired conducted power. The undesired conducted power shall be in accordance with EIA RS-204-C-82.

3.2.3.12 Undesired radiated power. The undesired radiated power shall be in accordance with EIA RS-204-C-82.

3.2.3.13 Frequency stability. Frequency stability shall be within ±0.001 percent of the assigned frequency over a minimum of 10 hours when tested to the limits specified in 3.2.1, 3.4.1, and 3.4.2.

3.2.3.14 Remote output. The output power shall be 0 dBm into 600 ohms.

3.2.3.15 Audio output impedance. The audio output impedance shall be 600 ohms balanced ±20 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

3.2.4 Functional control performance requirements. Functional control performance requirements shall be as specified in 3.2.4.1 and 3.2.4.2.

3.2.4.1 Controls. The control unit shall have, as a minimum, the functional switches and controls specified in a through g:

a. Speaker volume level control (one for each receiver)
b. Squelch sensitivity control (one for each receiver)
c. Channel selector
d. High/low RF power switch
e. AC power ON/OFF switch (AC power switch may be located on the R/T unit)
f. Mode switch (US/International)
g. Dimmer switch

3.2.4.2 Indicators. The control unit shall have, as a minimum, the functional illuminated indicators specified in a through d:

a. AC power ON
b. RF power ON
c. Channel selected
d. BIT and BITE fault indicator

3.2.5 Antenna. The R/T unit shall be designed to operate from a single marine band antenna.

3.2.6 Enclosure. The degree of enclosure shall be splashproof construction in accordance with MIL-STD-108.

3.3 Assembly and subassembly interchangeability. Mechanical and electrical interchangeability shall exist between identically numbered assemblies and subassemblies of the equipment. Provisions shall be made in the design tolerance to accommodate various solid-state components, such as microcircuits and other parts, without departure from specified performance. No degradation in performance limits shall be permitted as a result of interchangeability requirements.

3.4 Environmental conditions. Environmental conditions shall be as specified in 3.4.1 through 3.4.6.

3.4.1 Temperature range. The equipment shall be operable over an ambient temperature range of 0° Celsius (C) to +50°C with the maximum degradations specified in TABLE III.

Source: https://assist.dla.mil -- Downloaded: 202640-21T00:00Z
Check the source to verify that this is the currentversion before use.

TABLE III. Temperature range.

Equipment	Allowable degradation	Reference paragraph 1/
Transmitter		
Maximum power output	1 dB	3.2.2.1
FM hum and noise	6 dB	3.2.2.5
Carrier frequency stability	Within limits of 3.2.2.7	3.2.2.7
Receiver		
Usable sensitivity	3 dB	3.2.3.1
Audio squelch sensitivity	4 dB	3.2.3.3
Modulation acceptance bandwith	10 percent	3.2.3.4
Adjacent channel selectivity	8 dB	3.2.3.5
Audio power output	1 dB all outputs	3.2.3.8
Hum and noise ratio	8 dB	3.2.3.10
Frequency stability	Within limits of 3.2.3.13	3.2.3.13, 4.5.6.1

1/ Tests to be conducted in accordance with 4.5.3.

3.4.2 High humidity. High humidity shall be as specified in 3.4.2.1 and 3.4.2.2.

3.4.2.1 High humidity (receiver). The amount of receiver degradation shall be not greater than that specified in 3.4.1 and TABLE III when the receiver is tested in accordance with the High humidity test paragraph of EIA RS-204-C-82.

3.4.2.2 High humidity (transmitter). Inboard equipment shall maintain the specified performance when exposed to the humidity conditions specified in MIL-STD-810C, Method 507.2, Procedure III, except that testing shall be performed for a total of five cycles (120 hours).

3.4.3 Salt fog. The equipment shall comply with the requirement of the Salt fog (spray) paragraph of MIL-E-16400G, for sheltered equipment.

3.4.4 Vibration. The equipment shall conform to the vibration requirements of MIL-STD-167-1 for Type I, lightweight equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

3.4.5 Shock. The equipment shall conform to the shock requirements of MIL-S-901, Type A, Grade A (see 6.5.10), Class II, lightweight.

3.4.6 Inclination. The equipment shall be subjected to the Inclination test limits (except submarine installations) paragraph of MIL-E-16400G. The maximum angle shall be 60 degrees.

3.5 Handset. The equipment shall be designed to operate with a handset. Holder for the handset shall be provided along with the brackets to attach the holder to the control head front panel or on a flat surface located near the control head front panel.

3.6 Identification and marking. Identification and marking shall be in accordance with Identification and marking paragraph of MIL-E-16400G. Items with identification plates located in the equipment areas not readily visible shall have additional marking as specified in a through c:

a. Item name
b. Item nomenclature (if applicable)
c. Item serial number (if applicable)

This information shall be legibly and permanently marked by engraving, stamping, stenciling, silk screening, or other suitable means, and shall be visible without removing the item from the associated assembly.

3.6.1 ESDS nonstandard parts. ESDS nonstandard parts (see 6.5.8) shall be marked with the sensitive electronic device symbol specified in MIL-STD-129.

3.7 Reliability. Reliability shall be as specified in 3.7.1.

3.7.1 Quantitative reliability requirements. The equipment, including BITE, shall have an upper mean-time-between-failures (MTBF) (θ_0 as defined by MIL-STD-781) of 5400 hours, and a lower MTBF (θ_1 as defined by MIL-STD-781) of 2700 hours.

3.8 Maintainability. Maintainability shall be as specified in 3.8.1 and 3.8.2.

3.8.1 Qualitative maintainability requirements. The equipment shall provide ease of maintenance, accessibility, and replacement of all modules and parts.

3.8.2 Quantitative corrective maintenance. The equipment, including BIT, shall have a mean-corrective-maintenance-time (M_{ct}) not exceeding 0.5 hour and a maximum-corrective-maintenance-time (95th percentile) not exceeding 1.0 hour when corrective maintenance is accomplished at the organizational level of maintenance by

Source: https://assist.dla.mil -- Downloaded: 202640-21T00:00Z
Check the source to verify that this is the currentversion before use.

the replacement of lowest subassemblies (modules, printed circuit boards) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. BIT circuits are included in the corrective maintenance time requirement. The equipment is considered to be in a corrective maintenance mode whenever a failure (see 6.5.3) or malfunction occurs, including a malfunction that allows degraded operation.

3.9 Workmanship. Workmanship shall be as specified in 3.9.1 and 3.9.2.

3.9.1 General workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

3.9.2 Random vibration. All equipment shall withstand a defect detection vibration screen of random vibration of $0.04g^2/Hz$ ±3 dB from 80 Hz to 350 Hz as specified in 4.8.2 and temperature cycling as specified in 4.8.2.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality and safety assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

measure quality and safety of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
    1. Production inspection (Group A) (see 4.4.1)
    2. Production control inspection (Group B) (see 4.4.2)
c. Inspection of packaging (see 4.9)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of the examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE IV.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on every equipment offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process, such as functional and performance tests at a limited number of points in the the required range, tests which detect deviations from design, and tests which detect hidden defects of materials. Production inspection shall consist of the examinations and tests specified in TABLE IV, Group A.

Source: https://assist.dla.mil -- Downloaded: 202640-21T00:00Z
Check the source to verify that this is the currentversion before use.

TABLE IV. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection	
				Group A	Group B
Dimensions	3.1.13.1	4.5.1	X	X	
Weight	3.1.13.2	4.5.1	X	X	
Parts, materials, and processes	3.1.3.1 through 3.1.3.1.1.6	4.5.1	X	X	
Identification and marking	3.6	4.5.14	X	X	
Safety	3.2.1.9	4.5.1	X		X
Design and construction	3.1.3 through 3.1.3.1.1.6, 3.1.4, 3.1.15, 3.2.1 through 3.2.6, 3.3, 3.5, 3.6, 3.6.1, 3.7, 3.8 through 3.8.2	4.5.1	X		
Environmental requirements:					
Temperature	3.4.1	4.5.3	X		X
Humidity	3.4.2	4.5.10	X		X
Salt fog	3.4.3	4.5.9	X		X
Vibration	3.4.4	4.5.7	X		X
Shock	3.4.5	4.5.8	X		X
Inclination	3.4.6	4.5.11	X		X
Receiver and guard receiver					
Usable sensitivity	3.2.3.1	4.5.6	X	X	

Source: https://assist.dla.mil - Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

TABLE IV. Examinations and tests - Continued.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection	
				Group A	Group B
Quieting sensitivity	3.2.3.2	4.5.6	X	X	
Modulation acceptance bandwidth	3.2.3.4	4.5.6	X	X	
Audio power output	3.2.3.8	4.5.6	X	X	
Hum and noise ratio	3.2.3.10	4.5.6	X	X	
Receiver adjacent channel selectivity and desensitization	3.2.3.5	4.5.6	X		X
Intermodulation spurious attenuation	3.2.3.7	4.5.6	X		X
Spurious response attenuation	3.2.3.6	4.5.6	X		X
Antenna overload	3.1.12	4.5.2	X		X
Power source	3.2.1	4.5.5, 4.5.6	X		X
Interchangeability	3.3	4.5.4	X		X
Controls and control circuits	3.1.4	4.5.13	X		X
Leakage current	3.2.1.8	4.5.12 through 4.5.12.2	X	X	
Enclosures	3.2.6	4.5.1	X		X
Reliability	3.7	4.6 through 4.6.1.2	X		
Maintainability	3.8	4.7 through 4.7.3	X		

Source: https://assist.dla.mil - Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

TABLE IV. Examinations and tests - Continued.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection	
				Group A	Group B
Operating test	3.2.2, 3.2.3, 3.2.4, 3.2.5, 3.4.1, 3.5	4.5.2	X	X	
Transmitter frequency accuracy	3.2.2.7	4.5.5	X	X	
RF power output (1 W and 25 W)	3.2.2.1	4.5.5	X	X	
Conducted spurious emissions	3.2.2.2	4.5.5	X		X
Audio frequency harmonic distortion	3.2.2.3	4.5.5	X	X	
FM hum and noise level	3.2.2.5	4.5.5	X	X	
Modulation limiting	3.2.2.6	4.5.5	X	X	
Modulation and noise spectrum	3.2.2.9	4.5.5	X	X	
Audio frequency response	3.2.2.4	4.5.5	X		X
Workmanship	3.9	4.8	X	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified herein, and shall consist of examinations and tests which encompass functional and performance tests throughout the entire range of operation, tests to determine deviations in the processes and materials required to fabricate the equipment, and environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole. One equipment shall be selected from each six consecutive production units for submission to Group B sampling tests. The equipment submitted for Group B sampling tests shall be selected by the procuring activity and shall be representative of current production. These tests shall be performed on the equipment as offered for delivery. Production control inspection shall include the examinations and tests specified in TABLE IV, Group B. Failure of the sample unit to conform to prescribed requirements shall constitute a failure and shall be cause for rejection of the sample unit. Group B inspection shall be performed on equipment that have passed Group A inspection specified in 4.4.1. The equipment shall conform to the requirements of production control inspection prior to release for shipment.

4.4.2.1 Rejected units. If an equipment is rejected, the contractor may withdraw the unit from further inspection. The contractor may also rework the rejected unit to correct the defect(s) and may reinspect the unit. Production control inspection for rejected units shall be as specified in 4.4.2. Rejected units shall be kept separate from new units and shall not lose their identity.

4.4.3 Nonconformance corrective action and retest. If a sample unit fails the inspection specified in 4.4.2, the contractor shall immediately investigate the cause of the failure and shall implement the corrective action necessary to correct the units of product which were manufactured under the same conditions, materials, processes, and so forth. The equipment then shall be retested.

4.4.4 Reinspection of conforming production control inspection sample units. Unless otherwise specified (see 6.2), sample units which have been subjected to, and have passed, production control inspection may be accepted on the contract, provided all damage has been repaired and the equipments are resubjected to, and pass, production inspection.

4.5 Test methods. The examinations and tests specified herein shall be performed as specified in 4.5.1 through 4.5.12.2.

4.5.1. Examinations and tests. Examinations and tests specified herein shall be conducted in accordance with the applicable inspection procedures of MIL-E-16400G and Government-approved test procedures. Equipment shall be examined for obvious defects of workmanship, assembly, fit, mechanical safety, and marking. The examination shall include verification of materials, parts, and finish used for

Source: https://assist.dla.mil -- Downloaded: 202610-21T00:00Z
Check the source to verify that this is the current version before use.

the equipment. It shall be the responsibility of the contractor to provide all cables, connectors, test instrumentation, test facilities, and other devices required to set up and perform all tests at the contractor's plant, or other place of test, to verify compliance with the requirements specified herein. When the equipment is tested as a system, the contractor shall provide interconnecting cables adequate to simulate realistic installation conditions; that is, length, attentuation, and shielding shall be as identical as possible to actual installations.

4.5.2 Operating test. The equipment shall be energized and subjected to an operating test to ensure quantitatively the proper functioning of the equipment including all operating controls (see 3.2.4) and accessories (see 3.2.5 and 3.5). The test shall include actual voice communications between the two equipments.

4.5.3 Temperature test. The parameters shall initially be measured at room temperature after the equipment has been stabilized. The equipment then shall be made inoperative and lowered in temperature to 0°C for a 2-hour soak period after which the equipment shall be turned on. Receiver measurements shall be made first, followed by the transmitter measurements. All measurements shall be completed within 1 hour. Following completion of the low temperature measurements, the temperature shall be raised to +50°C and allowed to soak for 8 hours while being operated at a 50 percent duty cycle of 1 minute transmit and 1 minute receive. Maximum RF power and audio power outputs shall be provided. Measurements shall be made following the soak period. No adjustments of equipment shall be permitted nor shall forced circulation of air be allowed directly on the equipment (see 3.4.1).

4.5.4 Interchangeability, examinations, and tests. The contractor shall select tests from this specification, exclusive of temperature, humidity, and vibration, to verify the interchangeability requirements of 3.3. Examinations and tests shall be selected on the basis of being meaningful for verification of interchangeability to the particular assembly or subassembly being interchanged. Where mechanical criteria are not specified, the contractor shall provide sufficient detail to the test methods to verify the mechanical aspects of interchangeability.

4.5.5 Transmitter tests. The transmitter shall be tested in accordance with the Method of measurement paragraphs of EIA RS-152-C-88, except for temperature tests.

4.5.6 Receiver tests. The receivers shall be tested in accordance with the Method of measurement paragraphs of EIA RS-204-C-82, except for temperature tests.

4.5.6.1 Frequency stability. Frequency stability measurements shall be made by measuring the receiver operating frequency while maintaining the last intermediate frequency constant.

Source: https://assist.dla.mil -- Downloaded: 202640-21T00:00Z
Check the source to verify that this is the current version before use.

4.5.7 Vibration. The equipment, less antenna, shall be vibrated to determine conformance to the requirements of 3.4.4, without shock mounts.

4.5.8 Shock. The equipment, less antenna, shall be shock tested to determine conformance to the requirements of 3.4.5.

4.5.9 Salt fog test. The equipment shall be subjected to the salt fog test specified in MIL-STD-810C, Method 509.1, Procedure I, except as otherwise specified herein.

4.5.10 High humidity. The equipment shall be tested to determine conformance to the requirements of 3.4.2.

4.5.11 Inclination test. The equipment shall be tested to determine conformance to the requirement of 3.4.6.

4.5.12 Leakage current test. Leakage current test shall be performed as specified in 4.5.12.1 and 4.5.12.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.12.1 Equipment connections. Leakage current shall be measured on equipment in its normal operating configuration. Each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment under test and the source power shall be opened. If the equipment must be touched during the test, personal protective devices, (for example, rubber gloves, insulating tools, or knobs), shall be used for protection against electric shock. The equipment shall be connected as shown on FIGURE 1. Measurement under DC power source condition is not required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

4.5.12.2 Measurement. The leakage current shall be determined by the voltage-drop method. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 volts (V) at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. The overall measurement error shall not exceed 5 percent. A true rms voltmeter shall be used. The probe shall be connected to the equipment enclosure and associated unit enclosures and measurements taken for every combination of switch positions available on FIGURE 1. The probe also shall be connected to all external conducting parts, such as, connector housings, case, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available as shown on FIGURE 1. Equipment having more than one mode of operation, such as transmit and receive, shall be tested in each mode. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.13 Controls and control circuits tests. The equipment shall be tested to determine the suitability of controls and control circuits for satisfactory mechanical and electrical operation in accordance with 3.1.4, 3.1.5, 3.1.6, 3.1.14, 3.1.15, and 3.2.4.

4.5.14 Identification and marking. Examination shall be performed to determine conformance to the requirements of 3.6 and 3.6.1.

4.6 Reliability testing. A sample of at least three equipment (excluding antenna and handset) shall be selected from the first month's production for reliability qualification testing in accordance with Test Plan IVD of MIL-HDBK-781. Environmental and operational duty cycles shall be as specified herein. The reliability test shall be successfully completed prior to delivery of any equipment under the contract.

4.6.1 Temperature cycling. Temperature cycling between 0°C (+3°C, -0°C) and +50°C (+0°C, -3°C) shall be performed throughout the test with the constraints specified in a and b:

a. Time duration at the stabilized high and low temperatures shall be approximately 3 hours.

b. During transition between temperature extremes, the average rate of chamber temperature change shall be not less than 5°C per minute.

4.6.1.1 Rejected equipment. When equipment has been rejected as a result of the reliability test, the whole lot from which the equipment was drawn shall be reworked as necessary, reinspected, and another random sample shall be drawn from another lot for another reliability test. In the event that the cause of failure is the equipment manufacturers' design or the part manufacturers' designs, corrective action shall be performed on all equipment delivered under the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

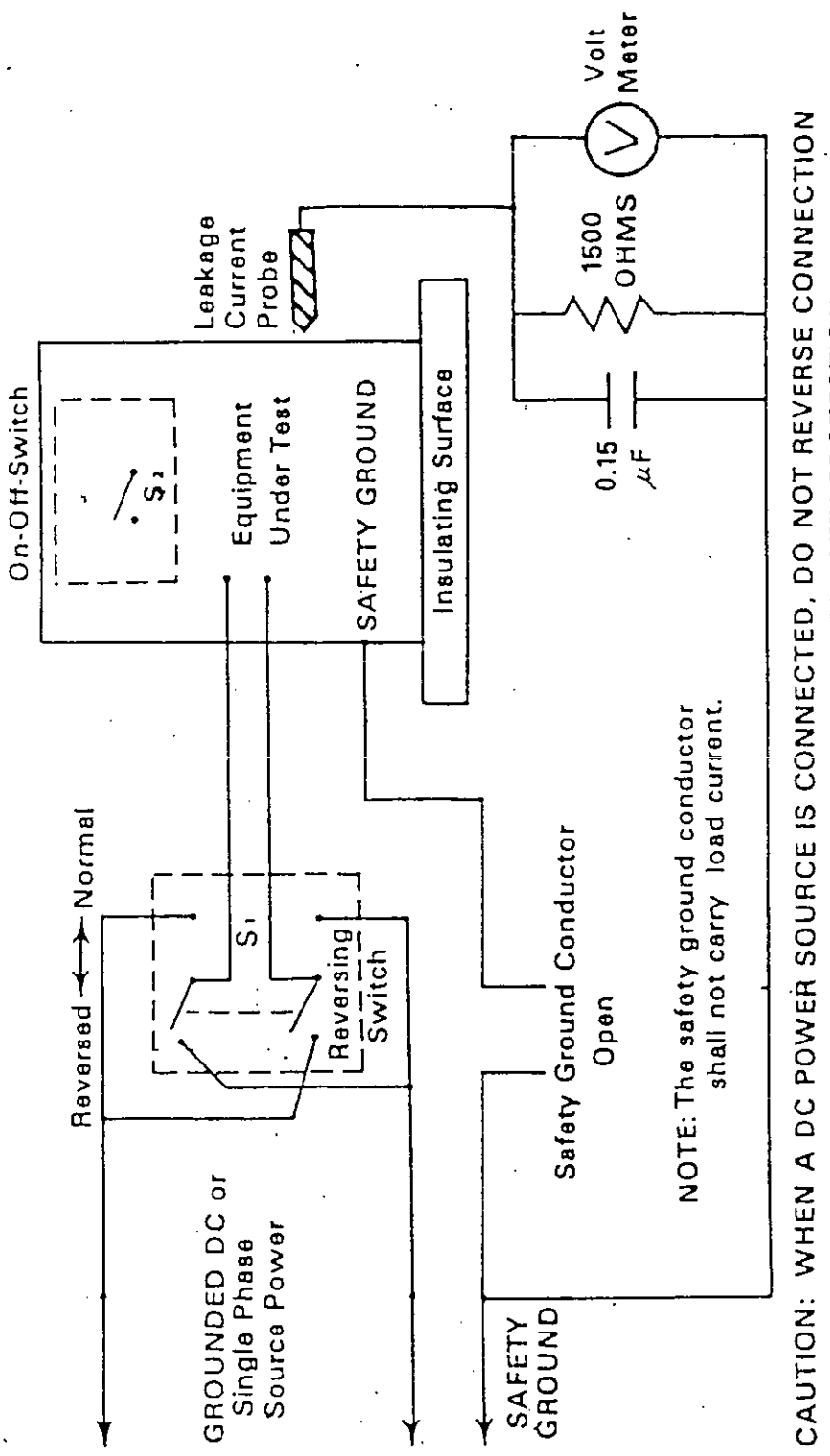


FIGURE 1. Single-phase test diagram for leakage current.

Source: https://assist.dla.mil -- Downloaded: 202610-21T00:00Z
Check the source to verify that this is the current version before use.

4.6.1.2 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and reconditioning as required and shall be subjected to the production tests of 4.4.1 to ensure that the equipment conforms to the requirements of this specification. After reconditioning, reinspection, and retesting, the equipment may be delivered as part of the contract quantity, provided the equipment conforms to all terms and conditions of the criteria for acceptance.

4.7 Maintainability demonstration. The first article radio set shall be subjected to a maintenance capability and diagnostics demonstration at the organizational level. The demonstration shall be performed by a qualified technician and shall verify conformance to the M_{ct} requirements specified in 3.8.

4.7.1 Demonstration conditions. The demonstration shall be conducted in accordance with MIL-STD-471, Test Method 9.

4.7.2 Demonstration environment. The demonstration shall be performed in an environment similar to that in which the equipment will be installed.

4.7.3 Acceptance or rejection. The maintainability demonstration shall exhibit that the equipment conforms to the accept criteria prior to delivery. Failure to conform to the requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies
b. Upon approval of an approach, the contractor shall repeat the demonstration until an accept decision is reached.

4.8 Workmanship. Workmanship shall be as specified in 4.8.1 and 4.8.2.

4.8.1 General workmanship. The antenna shall be examined for conformance to the requirements of 3.9.1.

4.8.2 Random vibration. Each equipment shall be subjected to vibration, in turn, along each of three rectilinear orientation axes of the equipment: vertical, side-to-side, and front-to-back. Two or three axes of simultaneous vibration also may be used. A vibration survey shall be performed at an amplitude lower than that shown on FIGURE 2 to determine that no damaging levels of response are present. The spectrum may require amplitude notching to successfully perform the vibration screen at the levels shown on FIGURE 2. The contractor shall inform the procuring activity

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

of required changes to FIGURE 2. The vibration shall be performed at the end item level. All the hardware, including clamps and connectors, shall be exposed to vibration to determine conformance to the requirements of 3.9.2. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in each axis. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 2. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at test points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analyses of the acceleration spectral density (ASD). The instantaneous acceleration peaks may be limited to three times the rms acceleration level. The item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during vibration shall be corrected and the vibration continued for a cumulative time of 10 minutes in each axis.

4.8.2.1 Temperature cycling. Each equipment shall be subjected to 10 cycles of the temperature curve shown on FIGURE 3. The temperature rate of change shall be not less than 5°C per minute. Equipment power shall be turned on and off at the indicated times. The equipment shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the equipment reaches 80 percent of the chamber temperature. When failures occur, the equipment shall be reworked and the cycling continued for a cumulative total of 10 cycles.

4.8.2.2 Equipment configuration. Individual equipment (as specified in 3.1.6) need not be tested in a system configuration provided provision is made to fully exercise all active electrical and electromechanical elements. The antenna and handset may be excluded from the test.

4.9 Inspection of packaging. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

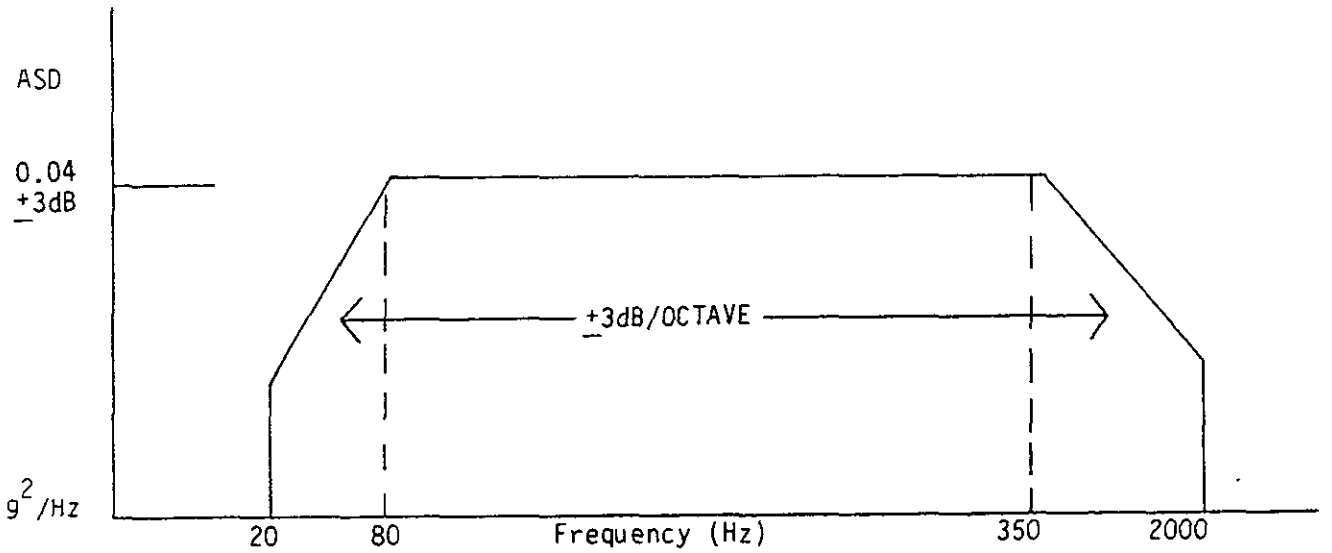


FIGURE 2. Random vibration curve.

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The equipment, covered by this specification, is intended for shipboard application to satisfy communications requirements in the Maritime Service in both national and international waterways.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. When reinspected production control sample units may not be accepted (see 4.4.4)
d. Level of preservation, packaging, packing, and marking required (see 5.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

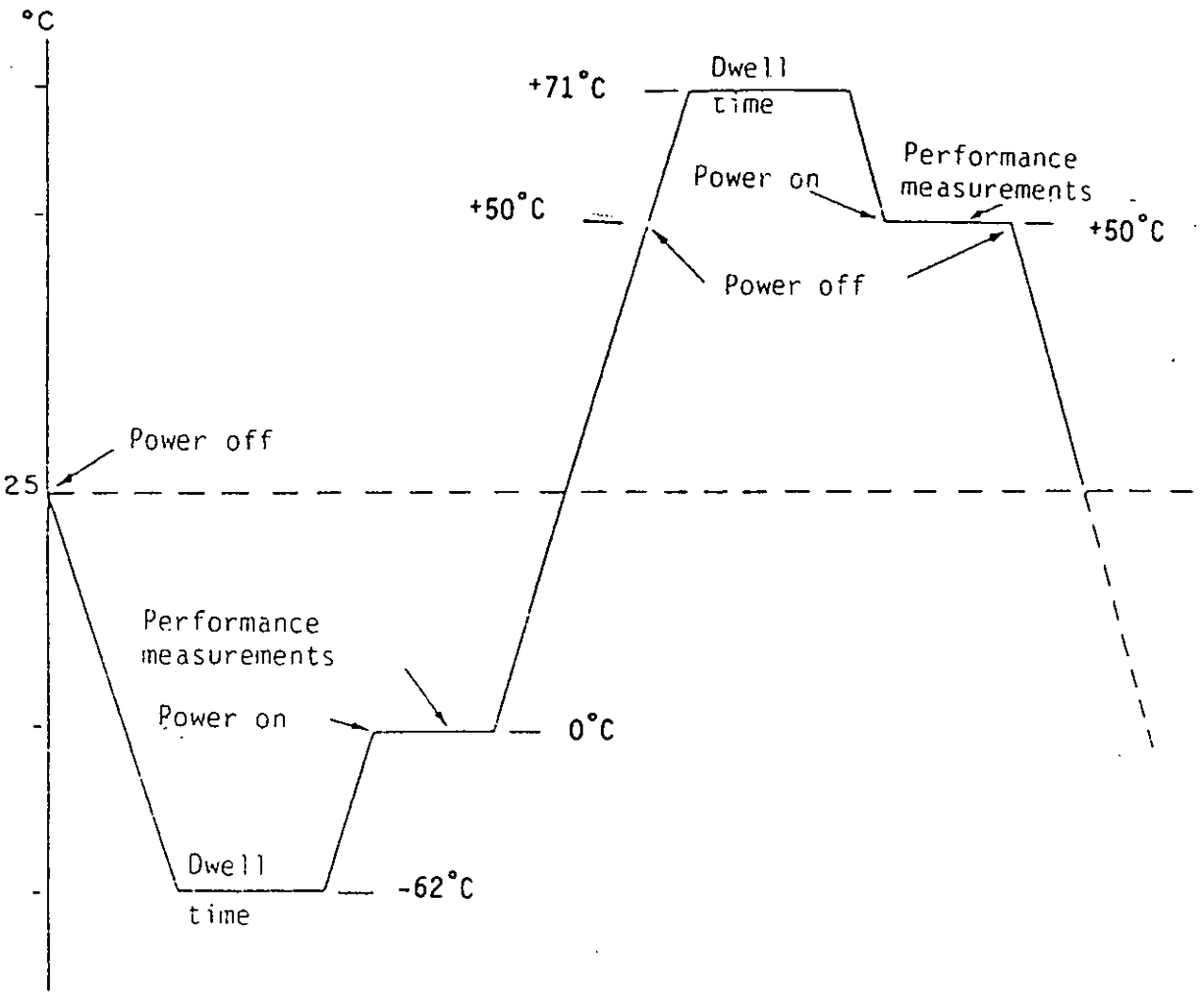


FIGURE 3. One cycle of temperature curve (not to scale).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

6.3 First article. When a first article inspection is required, the item(s) should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Periodic inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. Initial shipment of end items will be contingent upon successful completion of the periodic inspection and reliability testing.

6.5 Definitions. Definitions of terms used in this specification are given in 6.5.1 through 6.5.10.

6.5.1 Simplex mode. A simplex mode exists when the R/T of a R/T unit operate on the same frequency (f1/f1), but not simultaneously.

6.5.2 Half duplex mode. A half duplex mode exists when the R/T of a R/T unit operates on separate receive and transmit frequencies (f1/f2), but not simultaneously.

6.5.3 Failure. Any malfunction or parameter deviation which prevents the equipment from performing within the limits specified herein.

6.5.4 Assembly. A number of parts or subassemblies or any combination thereof, joined together to perform a specific function and capable of disassembly (such as, audio oscillator, amplifier, synthesizer, crystal oscillator).

6.5.5 BIT. BIT devices are an integral part of the equipment being tested. BIT may be automatic, manual, on-line, off-line, or a combination thereof.

6.5.6 BITE. BITE is functionally separate from, but connected to, the prime equipment and used for the express purpose of testing the prime equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

6.5.7 ESD. ESD is an abrupt momentary current flow between two objects upon the development of a low resistance current path between them. This current flow is the result of the equalization of different electrostatic potentials which can exist on electrically isolated objects prior to the development of low resistance current paths to other objects.

6.5.8 ESDS parts. ESDS parts are electrical and electronic parts (for example, transistors, diodes, microcircuits, resistors, capacitors, crystals) which are damaged when subjected to discharges up to 15,000 V from the human body. ESDS parts include all microelectronic and most semiconductor devices (with the exception of various power diodes, power transistors, and small signal zeners), thick and thin film resistors and chips, and devices and crystals.

6.5.9 ESD assemblies. All assemblies and the equipment cabinet or enclosure having one or more terminals, or exposed conduction paths connected directly or through a low impedance (not enough to protect the device) path to an ESDS part shall be considered as ESDS at that part.

6.5.10 Shock, Grade A. Shock, Grade A, items are defined as machinery, equipment, and systems essential for safety and continued combat capability of the ship. Shock testing is to demonstrate that the item will continue to perform its principal function without significant change in performance and that no portion of the equipment will come adrift or otherwise become a hazard to personnel or to Grade A machinery, equipment, or systems during and following application of shock loadings.

6.6 Subject term (key word) listing.

AN/URC-( ), Radio Set
Intended use: Shipboard and Maritime Service
Radio Set, AN/URC-( )

Preparing activity:
NAVY - EC

(Project No. 5820-N874(EC))

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:00Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

SPACE AND NAVAL WARFARE SYSTEMS COMMAND
WASHINGTON, D.C 20363-5100

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
SPACE AND NAVAL WARFARE SYSTEMS COMMAND
ATTN: PD 50P2F
WASHINGTON, DC 20363-5100

Source: https://assist.dla.mil -- Downloaded: 202610-21T00:00Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-29533	Radio Set, AN/URC-( )

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-21T00:00Z
Check the source to verify that this is the current version before use.